Networking Basics

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- Network Topologies
- Different types of Devices
- Types of ports, types of connections
- Understanding a network diagram
- DNS, NTP, HTTP, Wireshark

Lab

- Create a virtual network with Packet tracer
- Ping machines across networks
- Capture packets with Wireshark and ensure that you get the password

Network Topologies

- A network topology is the arrangement of devices (**nodes**) and connections (**links**) in a computer network.
- It shows how computers, servers, and other devices are connected and how data flows between them.
- There are two main types of topology:
  - **Physical Topology**:
    - The actual physical layout of cables and devices.
  - **Logical Topology**:
    - How data moves across the network, regardless of physical layout.

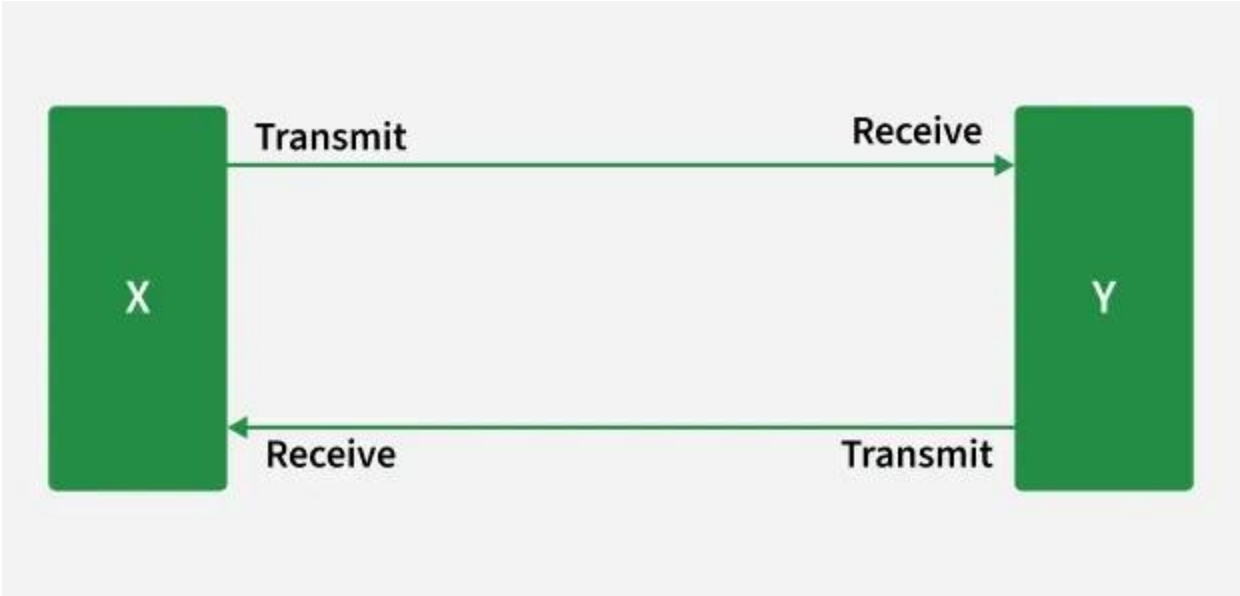
Network Topologies

Various networking topologies:

- Point To Point Topology
- Mesh Topology
- Star Topology
- Bus Topology
- Ring Topology
- Tree Topology
- Hybrid Topology

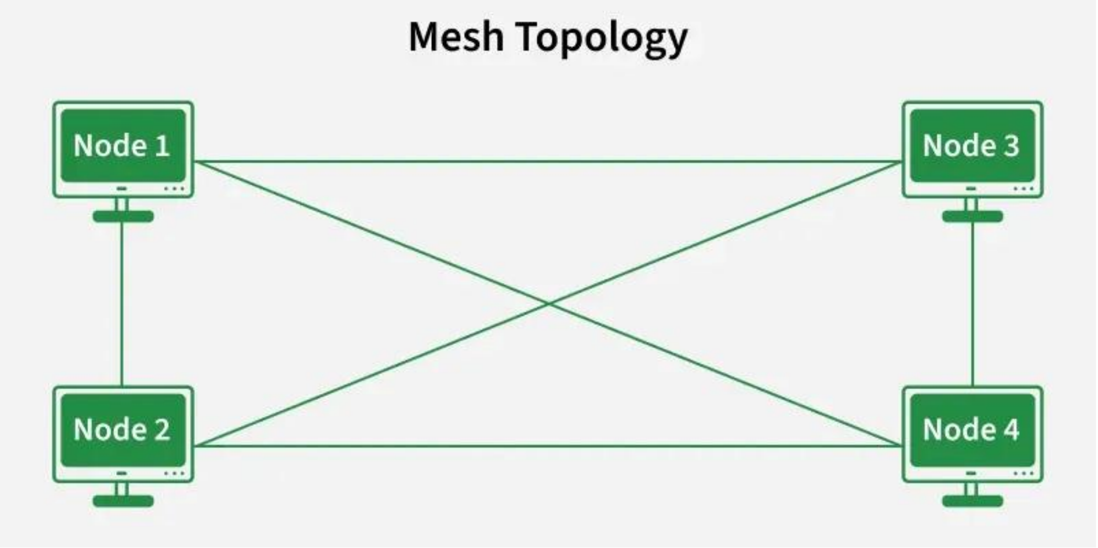
Point To Point Topology

- Point-to-point topology is a type of topology that works on the functionality of the sender and receiver.
- It is the simplest communication between two nodes, in which one is the sender and the other one is the receiver.
- Point-to-Point provides high bandwidth.



Mesh Topology

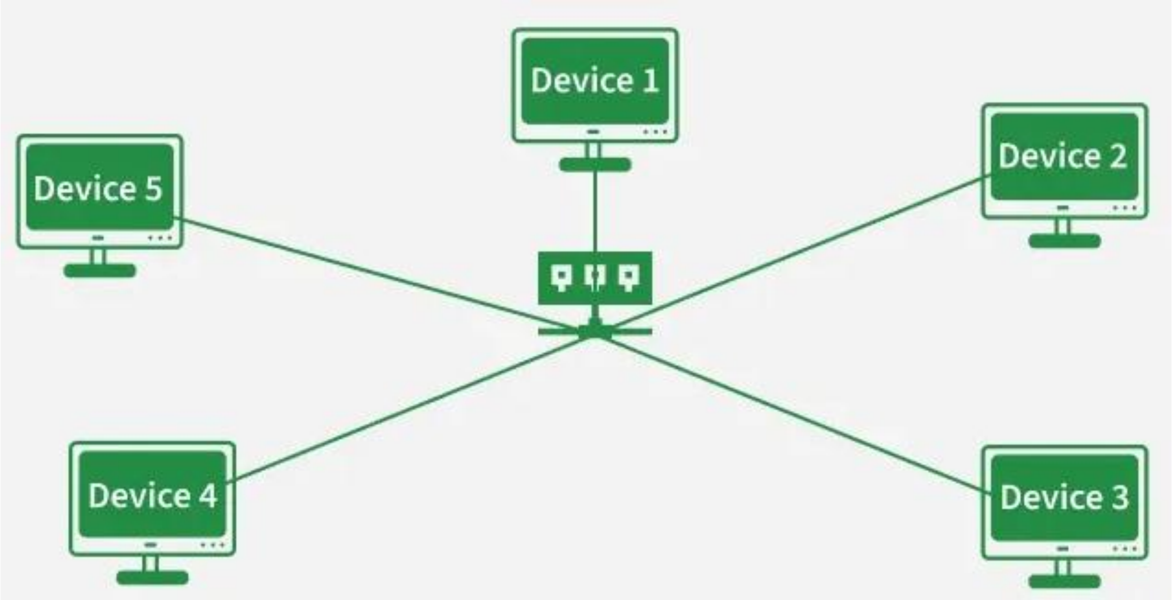
- In a mesh topology, every device is connected to another device via a particular channel. Every device is connected to another via dedicated channels.
- These channels are known as links.
- Communication is very fast between the nodes. Mesh Topology is robust.



- The fault is diagnosed easily. Data is reliable because data is transferred among the devices through dedicated channels or links.
- Provides security and privacy.

Star Topology

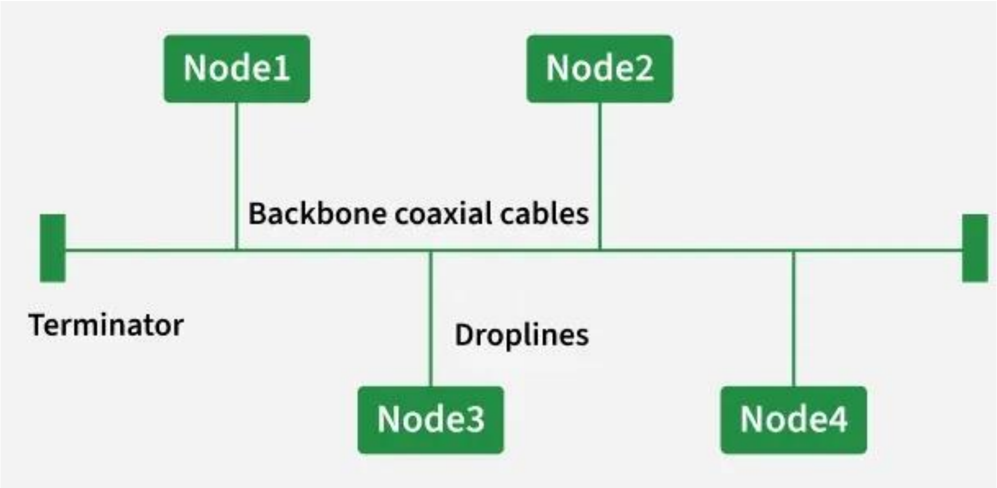
- In Star Topology, all the devices are connected to a single hub through a cable.
- This hub is the central node and all other nodes are connected to the central node.



- The hub can be passive in nature i.e., not an intelligent hub such as broadcasting devices, at the same time the hub can be intelligent known as an active hub.
- Active hubs have repeaters in them.

Bus Topology

- Bus Topology is a network type in which every computer and network device is connected to a single cable.
- It is bi-directional. It is a multi-point connection and a non-robust topology because if the backbone fails the topology crashes.

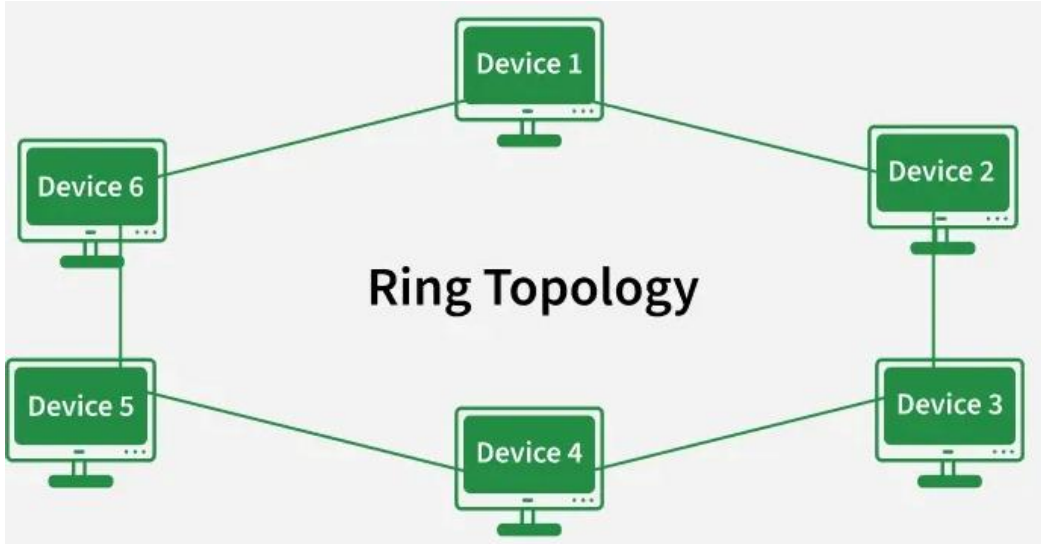


- If N devices are connected to each other in a bus topology, then the number of cables required to connect them is 1, known as backbone cable, and N drop lines are required.
- Coaxial or twisted pair cables are mainly used in bus-based networks that support up to 10 Mbps.

Ring Topology

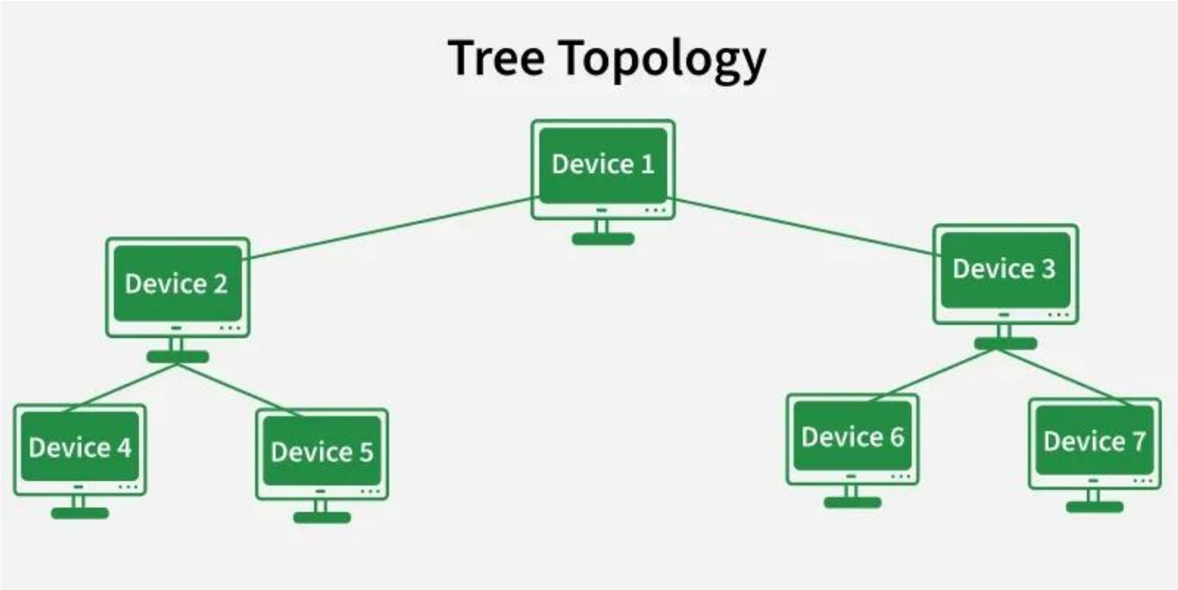
- It forms a ring connecting devices with exactly two neighboring devices.

- A number of repeaters are used for Ring topology with a large number of nodes, because if someone wants to send some data to the last node in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node.



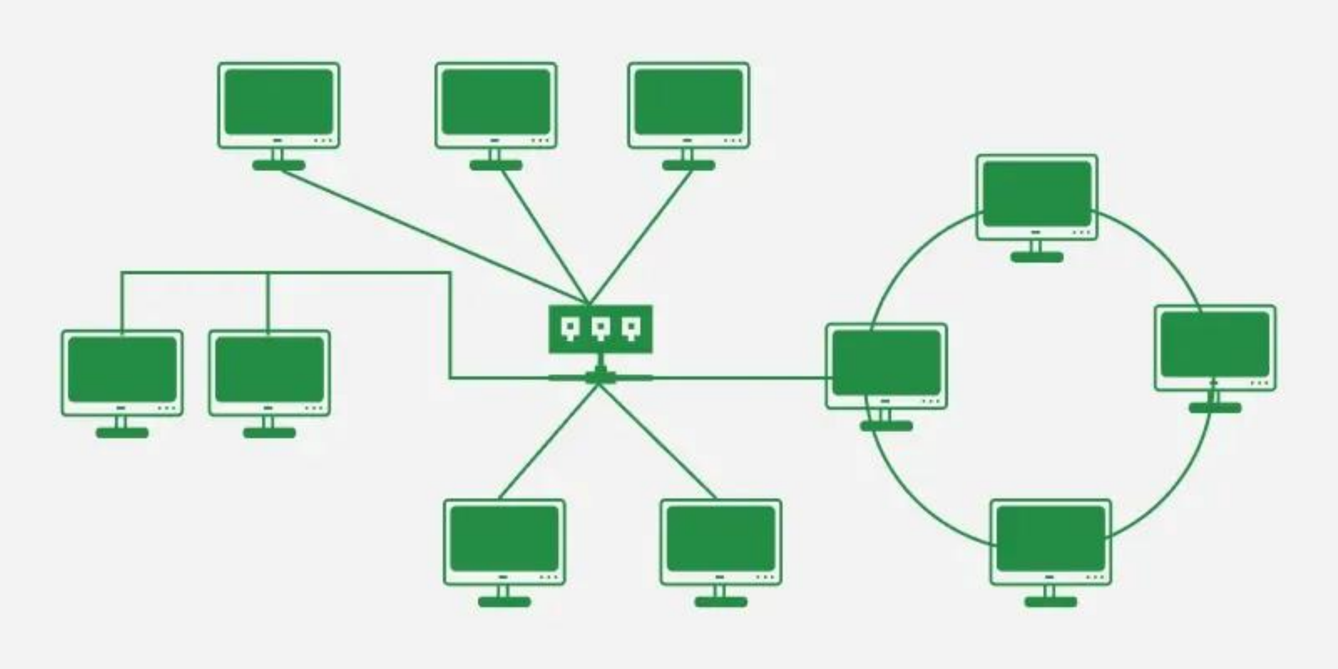
- In-Ring Topology, the Token Ring Passing protocol is used by the workstations to transmit the data where, Token passing is a network access method in which a token is passed from one node to another node & Token is a frame that circulates around the network.

Tree Topology



- Tree topology is the variation of the Star topology. This topology has a hierarchical flow of data.
- It allows more devices to be attached to a single central hub thus it decreases the distance that is traveled by the signal to come to the devices.
- It allows the network to get isolated and also prioritize from different computers.
- We can add new devices to the existing network.

Hybrid Topology



- Hybrid Topology is the combination of all the various types of topologies we have studied above.
- Hybrid Topology is used when the nodes are free to take any form.
- It means these can be individuals such as Ring or Star topology or can be a combination of various types of topologies seen above.

Start the packet analysis using Wireshark

tcp.stream eq 0

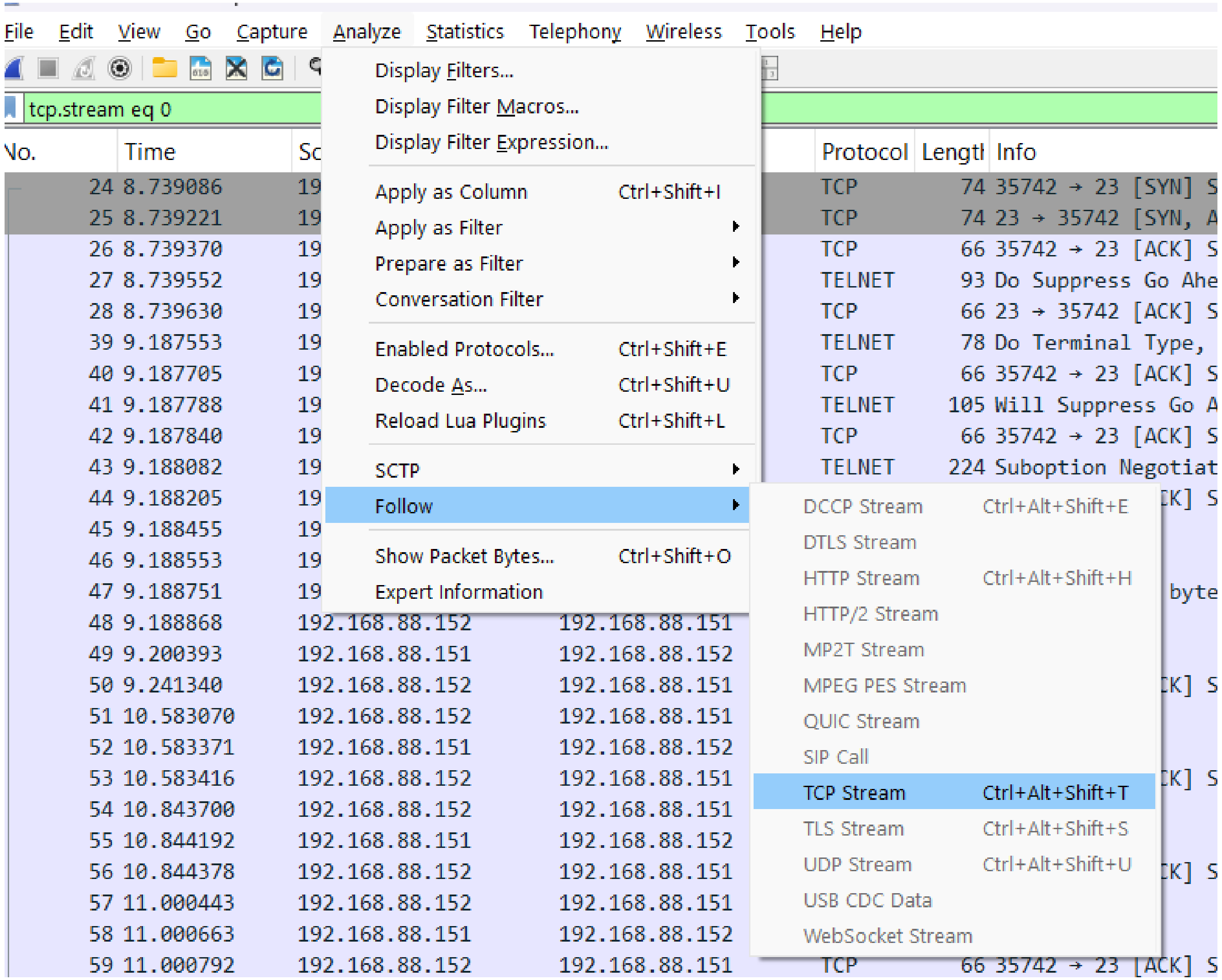
No.	Time	Source	Destination	Protocol	Lengtl	Info
24	8.739086	192.168.88.152	192.168.88.151	TCP	74	35742 → 23 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3266250721 TSecr=0 WS=128
25	8.739221	192.168.88.151	192.168.88.152	TCP	74	23 → 35742 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=366643550 TSecr=326
26	8.739370	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3266250721 TSecr=366643550
27	8.739552	192.168.88.152	192.168.88.151	TELNET	93	Do Suppress Go Ahead, Will Terminal Type, Will Negotiate About Window Size, Will Terminal Spee
28	8.739630	192.168.88.151	192.168.88.152	TCP	66	23 → 35742 [ACK] Seq=1 Ack=28 Win=65152 Len=0 TSval=366643550 TSecr=3266250721
39	9.187553	192.168.88.151	192.168.88.152	TELNET	78	Do Terminal Type, Do Terminal Speed, Do X Display Location, Do New Environment Option
40	9.187705	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=28 Ack=13 Win=64256 Len=0 TSval=3266251169 TSecr=366643998
41	9.187788	192.168.88.151	192.168.88.152	TELNET	105	Will Suppress Go Ahead, Do Negotiate About Window Size, Do Remote Flow Control, Don't Linemode
42	9.187840	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=28 Ack=52 Win=64256 Len=0 TSval=3266251169 TSecr=366643998
43	9.188082	192.168.88.152	192.168.88.151	TELNET	224	Suboption Negotiate About Window Size, Suboption Terminal Speed, Suboption X Display Location,
44	9.188205	192.168.88.151	192.168.88.152	TCP	66	23 → 35742 [ACK] Seq=52 Ack=186 Win=65024 Len=0 TSval=366643999 TSecr=3266251170
45	9.188455	192.168.88.151	192.168.88.152	TELNET	69	Do Echo
46	9.188553	192.168.88.152	192.168.88.151	TELNET	69	Won't Echo
47	9.188751	192.168.88.151	192.168.88.152	TELNET	114	Will Echo, 45 bytes data
48	9.188868	192.168.88.152	192.168.88.151	TELNET	69	Do Echo
49	9.200393	192.168.88.151	192.168.88.152	TELNET	80	14 bytes data
50	9.241340	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=192 Ack=117 Win=64256 Len=0 TSval=3266251223 TSecr=366644011
51	10.583070	192.168.88.152	192.168.88.151	TELNET	67	1 byte data
52	10.583371	192.168.88.151	192.168.88.152	TELNET	67	1 byte data
53	10.583416	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=193 Ack=118 Win=64256 Len=0 TSval=3266252565 TSecr=366645394
54	10.843700	192.168.88.152	192.168.88.151	TELNET	67	1 byte data
55	10.844192	192.168.88.151	192.168.88.152	TELNET	67	1 byte data
56	10.844378	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=194 Ack=119 Win=64256 Len=0 TSval=3266252826 TSecr=366645655
57	11.000443	192.168.88.152	192.168.88.151	TELNET	67	1 byte data
58	11.000663	192.168.88.151	192.168.88.152	TELNET	67	1 byte data
59	11.000792	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=195 Ack=120 Win=64256 Len=0 TSval=3266252982 TSecr=366645811
60	11.200692	192.168.88.152	192.168.88.151	TELNET	67	1 byte data
61	11.200968	192.168.88.151	192.168.88.152	TELNET	67	1 byte data
62	11.201101	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=196 Ack=121 Win=64256 Len=0 TSval=3266253183 TSecr=366646011
63	11.710070	192.168.88.152	192.168.88.151	TELNET	68	2 bytes data

Connect through Telnet client to server

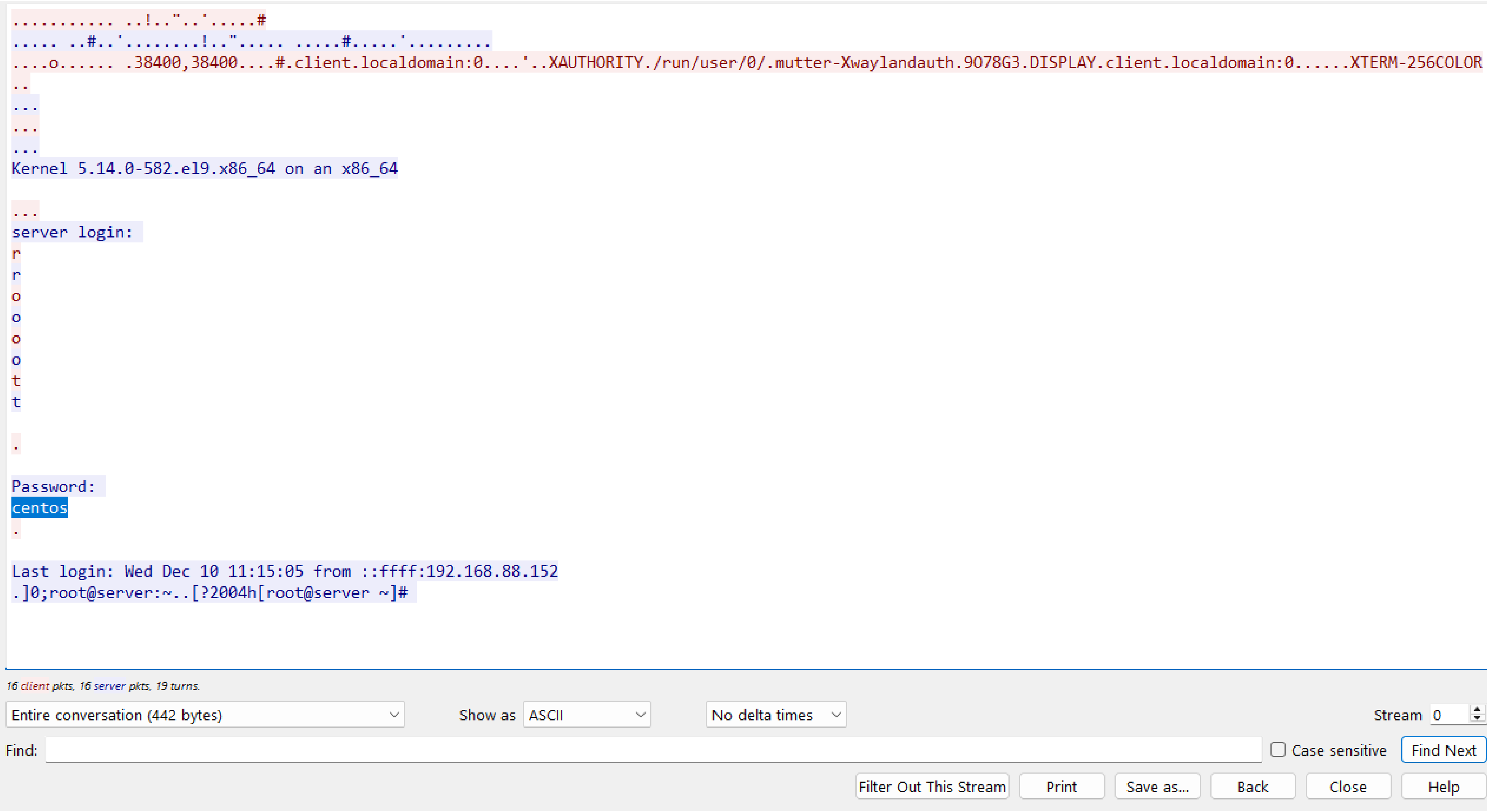
```
[root@client ~]# telnet 192.168.88.151
Trying 192.168.88.151...
Connected to 192.168.88.151.
Escape character is '^]'.

Kernel 5.14.0-582.el9.x86_64 on an x86_64
server login: root
Password:
Last login: Wed Dec 10 11:15:05 from ::ffff:192.168.88.152
[root@server ~]#
```

After logging in, stop capturing the traffic.

Analyze the captured packets



And you can see the server's Root password



Red & Blue

- Blue color-coding showing output displayed by terminal.

Wireshark · Follow TCP Stream (tcp.stream eq 0) · VMware Network Adapter VMnet8

```
..... ..#..'........!..".... .....#.....'..............
Kernel 5.14.0-582.el9.x86_64 on an x86_64
server login: root
Password:
Last login: Wed Dec 10 11:15:05 from ::ffff:192.168.88.152
.]0;root@server:~..[?2004h[root@server ~]#
```

- Red color-coding shows the input given to the terminal.

Wireshark · Follow TCP Stream (tcp.stream eq 0) · VMware Network Adapter VMnet8

```
........... ..!.."..'.....#....o...... .38400,38400....#.client.localdomain:0....'..XAU
THORITY./run/user/0/.mutter-Xwaylandauth.9O78G3.DISPLAY.client.localdomain:0......XTERM
-256COLOR........root
.centos
.
```


That's all Folks!